

CYX-AI: Local Action Prediction and Topology-Constrained Updates for Interactive Whole-Body PET/CT Lesion Segmentation

Yixin Chen, Wenpei Jiao, Yuyao Wang, and Xiangcen Wu

Institute of Medical Technology, Peking University Health Science Center, Beijing, China

Abstract. Interactive lesion segmentation should convert a corrective click into a targeted update without destabilizing an already strong automatic prediction. We present a method that retains a fixed five-model automatic PET/CT segmentation and performs subsequent corrections through two complementary local branches. A learned model predicts KEEP, ADD, and REMOVE actions from the images, the automatic mask, and cumulative clicks. In parallel, Topology-Aware Click Editing (TACE) derives a candidate from the automatic lesion-probability map by locally modulating its decision threshold. A sign-aware fusion rule combines the candidates, while component-split and new-hole checks reject structurally implausible updates and fall back to TACE or the previously accepted mask. We evaluated the correction stage with participant-grouped held-out folds on 1,607 studies from 1,274 participants. The protocol comprised six predictions under balanced centerline, random, and boundary click strategies. Normalized AUC-Dice was 0.5652 for the final initialization, 0.5850 for TACE, and 0.5964 for the fused policy. Among 1,038 studies with finite detection scores, AUC-DMM was 0.6759, 0.7076, and 0.7263. The fused policy exceeded TACE by 0.0114 AUC-Dice and 0.0186 AUC-DMM, with patient-cluster 95% confidence intervals excluding zero. Across 5,190 corrective interactions, the topology gate accepted no newly enclosed holes. These findings support bounded local correction under simulated interaction, while clinician-driven and hidden-test performance remain to be established.

Keywords: interactive segmentation · whole-body PET/CT · lesion segmentation · local editing · topology constraints

1 Introduction

Whole-body PET/CT combines metabolic and anatomical information for cancer assessment. Automatic lesion delineation remains difficult because uptake can be physiological or pathological, lesions vary markedly in size and location, and the appearance distribution changes across tracers and acquisition centres. The public FDG cohort [2] and PSMA cohort [5] expose this heterogeneity at scale. Strong multitracer systems can achieve accurate automatic masks [7], yet

even a high-quality initialization may contain a missed lesion, an over-segmented focus, or an incorrect boundary that requires human correction.

The autoPET V challenge studies this correction process directly. A system first predicts a lesion mask without user input and then receives cumulative foreground and background scribbles. Performance is summarized by the area under the interaction curve for voxelwise Dice and the connected-component detection matching metric (DMM), with equal weight assigned to the two metric ranks. The task therefore rewards both a strong initial mask and useful responses to later corrections. Because every prediction contributes to the area under the curve, an interactive method must improve the clicked error without unnecessarily altering regions that the user did not contest.

Mask guidance and focused updates have improved interactive segmentation in natural images [8,1]. Directly applying a global segmentation network after every click, however, leaves a mismatch between the local intent of the interaction and the spatial extent of the prediction. In whole-body lesion segmentation this mismatch is consequential. A background click near the centre of a large lesion can produce an artificial cavity, while an unconstrained foreground update can merge adjacent lesions and change component-level detection scores. The central methodological question is therefore not only how to use a click, but how to represent and accept a local edit while preserving the previously accepted prediction elsewhere.

We address this question by formulating each correction as a bounded action on the accepted mask. The proposed system combines a learned local action model with a training-free probability-based editor. A deterministic acceptance rule uses different evidence for additions and removals, rejects component-splitting removals, and prevents newly enclosed holes. We evaluate the correction stage on participant-grouped held-out folds. The governing question is whether the final fused system improves interaction AUC over its unchanged initialization and the preceding probability-based editor when structural rules restrict each correction. We separately test the invariant that no accepted update may introduce a newly enclosed background region.

2 Methods

2.1 Data and validation design

The source release contained 1,611 lesion-label volumes. Four FDG studies had neither a paired CT volume nor a paired PET volume and were excluded by a paired-input completeness rule. The analysis cohort therefore comprised 1,607 studies, including 1,010 FDG examinations from the FDG-PET-CT-Lesions collection [3,2] and 597 PSMA examinations from the PSMA-PET-CT-Lesions collection [6,5]. These studies represented 1,274 participants, of whom 245 contributed repeated examinations. Participant identity was defined by tracer and the de-identified participant identifier.

We used the challenge-provided `splits_final.json`. Validation folds contained 322, 321, 321, 321, and 322 studies. All examinations from one partici-

participant remained in a single fold, and no participant appeared in both the training and validation partitions of a fold. The learned correction model for each study was selected from that study’s held-out fold. The automatic model was a fixed five-model ensemble and was not re-estimated within these correction folds. Accordingly, our experiment estimates held-out-fold performance of the correction stage conditional on a fixed automatic initialization. It does not provide an independent estimate of automatic-model generalization.

The interaction simulator generated six predictions indexed from 0 to 5. The initial prediction used no scribble, followed by five cumulative corrections. Centerline, random, and boundary strategies were assigned to 536, 536, and 535 studies. For a metric sequence $q_0, \dots, q_5$, we report the normalized area under the interaction curve

$$\text{AUC}(q) = \frac{1}{5} \sum_{t=0}^4 \frac{q_t + q_{t+1}}{2}. \quad (1)$$

This normalization places AUC-Dice and AUC-DMM on the same scale as their per-prediction values.

2.2 Automatic initialization and state

The automatic initialization follows the five-model LesionTracer ensemble developed for multitracer whole-body PET/CT segmentation [7]. It uses a 3D residual-encoder nnU-Net architecture [4] and returns both a hard lesion mask M_0 and a float32 lesion-probability map P_0 . Under 18-neighbour connectivity, components smaller than 25 voxels are removed for FDG. For PSMA, components of at least 5 voxels are kept. A frozen logistic classifier retains 1–4-voxel components at probability 0.5 or higher using voxel count, physical volume, and bounding-box fill fraction. Because the challenge input does not expose a tracer label, a tracer router retains every second voxel along each axis of CT and PET. It maps non-finite values and intensities outside $[-2000, 10000]$ for CT or $[-5, 500]$ for PET to zero, and defines body voxels by $-500 < \text{CT} < 3000$. The router computes 99 summary features: 14 global PET features, five validity and axial-coverage features, six descriptors in each of 12 axial body bins, three axial uptake summaries, and five endpoint features. A histogram-based gradient boosting classifier, a standardized linear support vector machine, and a random forest each predict the tracer. PSMA is selected when at least two classifiers predict PSMA. Otherwise, FDG is selected. The result selects the tracer-specific automatic component filter, TACE parameters, and correction ensemble. We evaluated the classifiers using stratified, participant-grouped five-fold out-of-fold predictions on all 1,607 released studies from 1,274 participants. The majority vote was correct in all 1,607 out-of-fold studies, while individual classifiers disagreed on three studies. We then fitted each classifier to all released studies for deployment. The initial prediction is written without invoking either interactive branch.

For interaction $t > 0$, the system reads the previously accepted mask M_{t-1} and all accumulated clicks. A positive click is active only where it falls outside

M_{t-1} , and a negative click is active only where it falls inside M_{t-1} . A satisfied click does not independently open a new edit transaction, although all cumulative clicks remain input context for the learned branch. Both branches propose edits, but neither is permitted to replace the complete accepted mask without passing the local acceptance rules in Section 2.5.

2.3 Topology-Aware Click Editing

Topology-Aware Click Editing (TACE) is a training-free branch that converts clicks into local threshold changes on P_0 . Let G_+ and G_- denote the maximum-over-click Gaussian fields for positive and negative click cores. The local threshold field is

$$\tau_t(\mathbf{x}) = \text{clip}(0.5 - a_t G_+(\mathbf{x}) + r_t G_-(\mathbf{x}), 0, 1), \quad (2)$$

where a_t and r_t are increased when necessary to cross the median probability at an active click core. Each Gaussian has finite support ending at three standard deviations. A high-probability positive core that was removed by component filtering can drive the local threshold to zero. A low-evidence positive core receives a 3-mm minimum plateau, while removal has no minimum plateau. Positive edits remain fail-closed wherever $P_0 = 0$. The thresholded proposal is intersected with the active support and reduced to 26-connected, click-seeded components. Size limits are applied separately to each seeded component rather than to the complete mask. This procedure keeps voxels outside the support bit-identical to M_{t-1} .

TACE applies sign-specific topology rules before returning a candidate. Additions may create one prompted component or extend one existing component, but are rejected when they bridge two existing lesions. Removals proceed from the lesion boundary inward and retain any voxel whose deletion would split the foreground under 18-neighbour connectivity. FDG uses positive and negative strengths of 0.25, Gaussian scales of 12 and 8 mm, and maximum component sizes of 20,000 and 10,000 voxels. PSMA uses strengths of 0.20 and 0.30, scales of 10 and 6 mm, and limits of 10,000 and 5,000 voxels.

2.4 Local action prediction

The learned branch predicts actions rather than a replacement segmentation. Its seven input channels are CT, PET, the immutable automatic mask M_0 , positive and negative click maps, and two click-support maps with a 12-voxel radius. Five fold-specific 3D residual-encoder nnU-Nets were trained separately for FDG and PSMA. Each out-of-fold study used only its held-out model, whereas the final container ensembles all five models for the detected tracer. Image-feature parameters are initialized from LesionTracer, while the new input parameters and three-class action head are initialized for the correction task. At every voxel the model predicts KEEP, ADD, or REMOVE. Legal action masking allows ADD only outside M_0 , REMOVE only inside M_0 , and either edit only within its click support.

Training samples were constructed from residual errors of M_0 . Every crop was forced to contain at least one such error. Each sample drew one to five cumulative corrections and one of the centerline, random, or boundary strategies, with both choices sampled uniformly. ADD targets were false-negative voxels reached by a positive click and lying inside its 12-voxel support. REMOVE targets were false-positive voxels reached by a negative click and lying inside its support. All remaining voxels were labelled KEEP. Cross-entropy used class weights 0.5, 2, and 2 for KEEP, ADD, and REMOVE, respectively.

Training optimizes the action labels and the mask reconstructed from the predicted edits. The objective is

$$\mathcal{L} = \mathcal{L}_{\text{CE}} + \mathcal{L}_{\text{Dice}}^{\text{action}} + \mathcal{L}_{\text{Dice}}^{\text{mask}} + 3\mathcal{L}_{\text{nondeg}} + 2\mathcal{L}_{\text{wrong}}, \quad (3)$$

where the non-degradation hinge penalizes a reconstructed mask that is worse than M_0 , and the wrong-edit term penalizes changing a voxel in the opposite direction to the target. At inference, action probability must exceed 0.8 and the winning edit must exceed KEEP by 0.1. Click-seeded action components larger than 2,000 voxels are rejected.

2.5 Sign-aware fusion and topology-constrained acceptance

The two candidates are reconciled against M_{t-1} rather than against an empty canvas. For active positive clicks, learned and TACE additions are combined by union to favour recovery of a missed focus. For active negative clicks, a voxel is removed only when both branches propose removal, apart from the clicked core itself. This agreement rule limits deletion of true lesion tissue. Candidate removals that divide one previous foreground component into multiple surviving components are rolled back.

The remaining candidate is inspected for background components that are newly enclosed by foreground. If such a hole appears, the complete fused edit is discarded and the system evaluates the TACE candidate. If TACE also introduces a new hole, the system returns M_{t-1} . Thus the accepted update can be written as

$$M_t = \mathcal{H}((M_{t-1} \cup A_t) \setminus R_t, M_t^{\text{TACE}}, M_{t-1}), \quad (4)$$

where A_t and R_t are the sign-aware fused actions and $\mathcal{H}$ is the ordered split and new-hole acceptance rule. An empty PSMA initialization bypasses the learned branch and uses the same TACE and previous-mask fallback sequence. Figure 1 summarizes the method and the observed interaction curves.

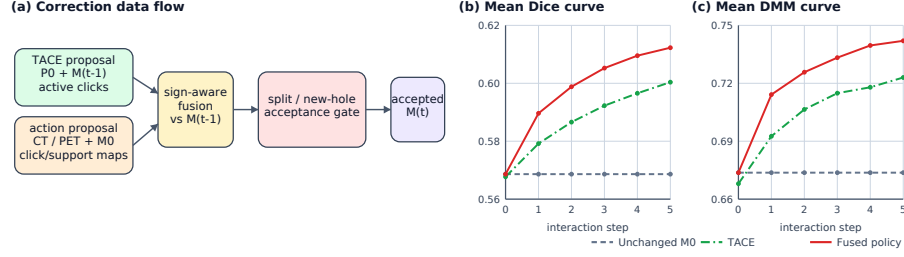


Fig. 1. Correction data flow and held-out-fold results for each system’s automatic initialization. (a) TACE uses P_0 , M_{t-1} , and active clicks, whereas local action prediction uses CT, PET, M_0 , and click/support maps. The ordered gate accepts the fused candidate or falls back to TACE and then M_{t-1} . (b,c) Curves show the mean at each prediction. Dice includes all 1,607 studies. DMM includes the 1,038 studies with finite values at all six predictions.

2.6 Training, inference, and statistical analysis

Each learned correction fold was trained for 150 epochs with 250 iterations per epoch, batch size 2, and $160 \times 224 \times 192$ voxel patches resampled to $3.0 \times 2.036 \times 2.036$ mm. CT followed nnU-Net normalization and PET was z-score normalized. Nesterov stochastic gradient descent used momentum 0.99, an initial learning rate of 3×10^{-4} , weight decay 3×10^{-5} , and polynomial decay. Training used mixed precision, deep supervision, a gradient norm limit of 1.0, Gaussian noise, blur, brightness and contrast changes, low-resolution simulation, gamma transformation, and mirroring. During training, an auxiliary best-checkpoint monitor used final-mask Dice minus twice the mean negative Dice change and 0.1 times the degradation rate. The reported out-of-fold evaluation used the final-epoch checkpoint from each study’s held-out fold. The final container ensembles all five final-epoch checkpoints for the detected tracer without test-time augmentation. The container was qualified for an NVIDIA T4 with 16 GiB memory and a 15-minute per-invocation limit. Out-of-fold evaluation used the released cohort tracer labels to select tracer-specific initialization filtering, TACE parameters, and the FDG or PSMA correction specialist. The TACE comparator retained its original component filter. The final fused candidate used the policy above. The reported interaction scores therefore do not incorporate routing error.

Branch latency was measured on the same node with eight allocated CPU cores and an NVIDIA L40 after P_0 and the interaction state were available. CPU-based TACE timing covered threshold modulation, component selection, and topology projection for 120 verified interactions from 24 cases, with three executions per state. The matched descriptive comparison used five click states from one FDG and one PSMA QA study. Local action prediction used the GPU and was executed once per state. Its timing included preprocessing, loading and initializing five packaged final-epoch checkpoints, five-fold sliding-window infer-

ence, action decoding, and inverse resampling. Both timings excluded automatic inference, Docker startup, input-file I/O, metrics, and fusion.

Primary comparisons were paired at the study level. Because 245 participants had repeated studies, confidence intervals were obtained from 10,000 participant-cluster bootstrap replicates. Each sampled participant contributed all eligible studies, while the estimand remained the study-level mean. We used percentile intervals and NumPy’s PCG64 generator with seeds 20260822 through 20260825 for the four prespecified comparisons. DMM is undefined for empty reference masks in the evaluation implementation. We therefore report DMM only for the 1,038 studies with finite values at all six predictions and do not impute undefined values. The frozen evaluator forms 18-connected components, counts a lesion as matched at intersection over union of at least 0.1, and does not penalize multiple assignments. We also counted final-prediction regressions relative to M_0 , candidate new-hole events, and the resulting fallback decisions.

Table 1. Algorithm details in the autoPET V reporting format.

Team name	algorithm name	data pre-processing	data post-processing	training data augmentation
CYX-AI	AutoPET V interactive lesion segmentation (ResEncL + EDT clicks)	nnU-Net CT normalization, PET z-score, resampling	Tracer-specific component filtering, TACE, sign-aware fusion, split and new-hole checks	Noise, blur, brightness, contrast, low resolution, gamma, mirroring
test time augmentation	ensembling	standardized framework	network architecture	loss
None	Five-model automatic ensemble and tracer-specific five-fold correction ensemble	nnU-Net v2 and PyTorch	3D residual-encoder U-Net with seven inputs and three action classes	Cross-entropy, action Dice, reconstructed-mask Dice, non-degradation hinge, wrong-edit penalty
training data	data/model dimensionality and size	use of pre-trained models	GPU hardware for training	
1,010 FDG and 597 PSMA paired PET/CT studies	3D, $160 \times 224 \times 192$ voxel patches	LesionTracer image-feature initialization, newly introduced action parameters initialized for this task	NVIDIA RTX 4090 and A800	

3 Results

All 1,607 studies received six predictions, with the planned domain, fold, and scribble-strategy counts. The final automatic initialization had a normalized AUC-Dice of 0.565181. The preceding TACE system yielded 0.584997, and the final fused policy yielded 0.596417. The paired gain of the fused policy over

its unchanged initialization was 0.031236, with a patient-cluster 95% confidence interval of 0.027155 to 0.035681. Its gain over TACE was 0.011420 with a confidence interval of 0.009256 to 0.013735. The mean Dice at the sixth prediction was 0.610474 for the fused policy compared with 0.598126 for TACE and 0.565181 for the final unchanged initialization.

Among the 1,038 finite-DMM studies from 855 participants, normalized AUC-DMM was 0.675877 for the final unchanged initialization, 0.707624 for TACE, and 0.726272 for the fused policy. Its paired gain was 0.050396 over the unchanged initialization, with a patient-cluster 95% confidence interval of 0.042447 to 0.059122. The gain over TACE was 0.018648 with a confidence interval of 0.014543 to 0.022986. Mean DMM at the sixth prediction reached 0.744143. For the fused policy, FDG yielded 0.590126 AUC-Dice and 0.716876 AUC-DMM, while PSMA yielded 0.607060 and 0.734971.

Table 2. Held-out-fold correction performance. Each row includes its associated automatic initialization. AUC values are normalized according to Eq. (1). DMM includes only studies with finite values at all six predictions.

Method	Studies	AUC-Dice	Dice at 5	Finite DMM	AUC-DMM
Final unchanged M_0	1,607	0.565181	0.565181	1,038	0.675877
TACE	1,607	0.584997	0.598126	1,038	0.707624
Fused correction policy	1,607	0.596417	0.610474	1,038	0.726272
FDG	1,010	0.590126	—	499	0.716876
PSMA	597	0.607060	—	539	0.734971

With P_0 available, TACE required no additional neural-network forward pass. The matched median-latency ratio for local action prediction versus TACE was 58.4 (Table 3). Across 120 TACE interactions, median, 95th-percentile, and maximum latencies were 0.589, 1.926, and 2.229 s.

Table 3. Per-click branch latency for ten matched states from one FDG and one PSMA study on the same node. TACE ran on CPU, and local action prediction used an NVIDIA L40. Values are seconds. IQR denotes interquartile range, and P95 denotes the 95th percentile.

Branch	Loaded checkpoints	Median (IQR)	P95
TACE	0	1.008 (0.336–1.866)	2.084
Local action prediction	5	58.843 (51.602–66.181)	68.208

Across 5,190 corrective interactions with non-empty reference masks, the new-hole check identified 1,090 fused proposals that would have introduced at least one newly enclosed background voxel. It selected TACE in 259 interactions and returned the previous mask in 831. No newly enclosed hole was accepted. This structural guarantee did not imply per-case metric monotonicity. Final Dice was

below its initial value in 18 of 1,607 studies, and final finite DMM was below its initial value in 44 of 1,038 studies.

4 Discussion

The fused policy improved both voxel overlap and component-level detection over the final initialization and TACE, with paired patient-cluster confidence intervals excluding zero. Agreement between these metrics matters because a small voxelwise edit can create, remove, split, or merge a detected lesion. The curves show gains after successive interactions, but the TACE comparison also includes the final component-filter update and does not isolate fusion.

The two branches encode different evidence. TACE edits the fixed probability landscape, while the learned branch uses image context and cumulative clicks to predict a larger action. The system unions plausible additions but requires agreement for removals because removing true lesion tissue can reduce both metrics. The experiment supports the fused policy as a whole. It does not isolate each rule or prove tracer specialization superior to a matched unified model. With cached P_0 , TACE supplies the lower-latency fallback without another neural-network forward pass, although the benchmark excludes complete container execution.

The topology gate addresses a concrete failure mode rather than an abstract claim of universal safety. More than one thousand fused candidates triggered the new-hole test, and the ordered fallback prevented every such candidate from being accepted. This result supports the implemented invariant within the evaluated interactions. It does not establish monotonic metric improvement, full topology preservation, or immunity to inaccurate human input. The observed case-level regressions show why the distinction is necessary.

Several limitations bound the interpretation. First, clicks were generated from reference-mask errors and do not reproduce clinician intent, imprecision, correction time, or disagreement. Second, the held-out folds apply to the learned correction models, while M_0 comes from a fixed five-model ensemble. The experiment therefore evaluates correction conditional on this initializer and is not strict out-of-fold validation of the entire pipeline. Router performance was evaluated only within the released cohort. Its 1,607/1,607 participant-grouped out-of-fold result does not establish accuracy on the hidden tracer distribution. Interaction out-of-fold evaluation selected specialists from released tracer labels rather than from the deployed tracer router, so the reported correction scores exclude routing errors. Validation on hidden or external tracer distributions is therefore required. Third, DMM could be analyzed in only 1,038 studies because the remaining reference masks made the metric undefined. Fourth, no hidden Final test result was available at the time of writing. Finally, the selected structural checks control component splits and newly enclosed holes, but other shape errors remain possible.

5 Conclusion

The fused combination of local action prediction, probability-based editing, and topology-constrained acceptance improved both interaction AUC metrics over its unchanged mask and the preceding TACE system without accepting newly enclosed holes. Clinician-driven evaluation is required to determine whether these gains transfer to real correction behaviour.

Code availability

Source code, container instructions, frozen configuration, checksums, and links to the public model-weight release are available in the main code repository. The frozen weights are provided in the separate weights-v0.3.0 release.

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